Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18EEOE01]

SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER–2021

RENEWABLE ENERGY SOURCES

Time: 3 Hours **Maximum Marks: 70**

Instructions:
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
- **ii.** Missing data may be suitably assumed.

1.	**a.**	Which are the different types of solar thermal collectors? Explain any one type in detail.	**07**
	b.	Explain in detail the P – V and I – V characteristics of PV cell.	**07**
2.	**a.**	What are the factors affecting wind motion? Explain in detail each factor.	**05**
	b.	Write a short note on offshore wind energy.	**05**
	c.	Mention few applications of wind energy.	**04**
3.	**a.**	Mention the advantages and limitations of tidal energy.	**06**
	b.	With necessary block diagram explain open OTEC cycle, write TS diagram also.	**08**
		OR	
4.	**a.**	Mention the advantages and disadvantages of wave energy.	**05**
	b.	Explain how energy is extracted from geothermal energy.	**04**
	c.	Explain any one process of biomass energy conversion.	**05**
5.	**a.**	Write short note for the following topics **i.** Hydrogen energy **ii.** Biodiesel production	**10**
	b.	State the disadvantages of fuel cells.	**04**
		OR	
6.	**a.**	What are fuel cells? How are fuel cells different from batteries?	**04**
	b.	What is MHD generation? Explain in detail the working with necessary diagrams.	**10**
7.	**a.**	Differentiate between sensible heat and latent heat in thermal energy storage.	**06**
	b.	Explain the operation principle of flywheels.	**08**
		OR	
8.	**a.**	Explain in detail Electrical energy battery storage systems with necessary diagrams.	**06**
	b.	Define energy storage. Explain the need for energy storage. Mention the different methods of energy storage.	**08**